PHYDSC402T — Classical Mechanics

Unit 1: Central Force Motion

7th Semester, Assam University (FYUG) — Haflong Government College

Paper: PHYDSC402T, Full Marks 100 [ESE 70 + CCA 30] | Unit 1: 12 lectures

Notes 8 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

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Unit syllabus (as in the syllabus file)

1. Central Force and its properties, conservative nature of Central Force, central orbit, differential equation of motion of a particle moving under central force in plane polar co-ordinate system.
2. Two-body central force problem: reduction to one body problem, equations of motion, Deduction of Kepler's laws of planetary motion and differential equation of the orbit.
3. Effective potential energy and radial motion, Stability of circular orbits, classification of orbits.

Top 2 high-mark topics in this unit

1. Deduction of Kepler's laws of planetary motion and differential equation of the orbit in plane polar coordinates — the core of the unit, a full multi-step derivation set as a long question in CBCS/MSc papers.
2. Effective potential energy, radial motion, stability of circular orbits and classification of orbits — derivation plus stability condition and orbit classification, named in the syllabus but with zero questions in any source checked, so high risk if set.

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1. Central Force and Its Properties

A central force is defined as a force acting on a particle such that its line of action always passes through a fixed point, known as the centre of force, and its magnitude depends solely on the distance r of the particle from that fixed point. Mathematically, a central force $\vec{F}(\vec{r})$ acting on a particle situated at a position vector $\vec{r}$ relative to the centre of force (chosen as the origin) can be expressed as

$$\vec{F}(\vec{r}) = F(r)\hat{r}$$

where $r = |\vec{r}|$ represents the magnitude of the position vector, $\hat{r} = \frac{\vec{r}}{r}$ is the unit vector along the radial direction pointing away from the origin, and $F(r)$ is a scalar function of distance. If $F(r) < 0$, the force acts towards the origin and is attractive; if $F(r) > 0$, the force acts away from the origin and is repulsive.

Physical examples of central forces include:

1. The gravitational force exerted by a point mass M on another point mass m, given by Newton's law of universal gravitation:

$$\vec{F}(\vec{r}) = -\frac{GMm}{r^2}\hat{r}$$

1. The electrostatic force exerted by a point charge q_1 on another point charge q_2, given by Coulomb's law:

$$\vec{F}(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\hat{r}$$

Conservation of Angular Momentum

The torque $\vec{\tau}$ acting on a particle subjected to a central force $\vec{F}(\vec{r})$ about the origin is defined by the cross product of the position vector and the force vector. Substituting the definition of a central force into the torque expression gives

$$\vec{\tau} = \vec{r} \times \vec{F}(\vec{r}) = \vec{r} \times \left(F(r)\frac{\vec{r}}{r} \right)$$

Using the property of the vector cross product that the cross product of any vector with itself is zero, we obtain

$$\vec{\tau} = \frac{F(r)}{r}(\vec{r} \times \vec{r}) = \vec{0}$$

By Newton's second law for rotational motion, the torque is equal to the time rate of change of angular momentum $\vec{L} = \vec{r} \times \vec{p}$, where $\vec{p} = m\vec{v}$ is the linear momentum of the particle:

$$\frac{d\vec{L}}{dt} = \vec{\tau}$$

Equating this derivative to the zero torque vector yields

$$\frac{d\vec{L}}{dt} = \vec{0}$$

Integrating with respect to time leads to the conservation law

$$\vec{L} = \text{constant vector}$$

Thus, under the action of a central force, the total angular momentum of the particle is conserved in both magnitude and direction.

Planar Nature of Motion

The conservation of the direction of angular momentum dictates the spatial geometry of the particle's trajectory. By definition, the angular momentum vector is orthogonal to the position vector $\vec{r}$ at every instant:

$$\vec{r} \cdot \vec{L} = \vec{r} \cdot (\vec{r} \times \vec{p}) = 0$$

Since $\vec{L}$ is a constant vector fixed in space, the condition $\vec{r} \cdot \vec{L} = 0$ implies that the position vector $\vec{r}(t)$ must always lie in a plane that passes through the centre of force and is perpendicular to $\vec{L}$. Consequently, the motion of a particle under a central force is entirely confined to a single plane.

Conservative Nature of Central Force

A force field is conservative if the work done by the force on a particle moving between two points is independent of the path taken. To show that a central force is conservative, consider a particle moving from an initial point $\vec{r}_1$ to a final point $\vec{r}_2$ along an arbitrary path C. The infinitesimal work done dW by the force during a displacement $d\vec{r}$ is

$$dW = \vec{F} \cdot d\vec{r}$$

Substituting $\vec{F} = F(r)\hat{r}$ into the work expression yields

$$dW = F(r)\hat{r} \cdot d\vec{r}$$

The displacement vector $d\vec{r}$ in spherical polar coordinates is given by $d\vec{r} = dr\hat{r} + rd\theta\hat{\theta} + r\sin\theta d\phi\hat{\phi}$. Evaluating the dot product with the radial unit vector $\hat{r}$ gives

$$\hat{r} \cdot d\vec{r} = dr$$

Therefore, the differential work done depends only on the change in radial distance:

$$dW = F(r)dr$$

Integrating from the initial radial distance r_1 to the final radial distance r_2, the total work done W_{12} along path C is

$$W_{12} = \int_{r_1}^{r_2} F(r)dr$$

This integral depends strictly on the initial distance r_1 and final distance r_2, proving that the work done is independent of the path connecting the two points. Hence, any central force is conservative.

Because the central force is conservative, there exists a single-valued scalar potential energy function $V(r)$ such that the force is the negative gradient of the potential energy:

$$\vec{F}(\vec{r}) = -\nabla V(r)$$

In spherical coordinates with radial symmetry, the gradient simplifies to

$$\vec{F}(\vec{r}) = -\frac{dV}{dr}\hat{r}$$

Equating the magnitude component with $F(r)$ yields the differential relationship

$$F(r) = -\frac{dV}{dr}$$

Integrating this relation from a reference radius r_0 (where $V(r_0) = 0$) to a distance r provides the potential energy function:

$$V(r) = -\int_{r_0}^{r} F(r')dr'$$

The work-energy theorem states that the net work done on the particle equals the change in its kinetic energy T :

$$W_{12} = T_2 - T_1$$

Substituting the potential energy definition into the work integral gives

$$W_{12} = \int_{r_1}^{r_2} \left(-\frac{dV}{dr}\right) dr = -[V(r_2) - V(r_1)] = V(r_1) - V(r_2)$$

Equating the two expressions for work done yields

$$T_2 - T_1 = V(r_1) - V(r_2)$$

Rearranging terms associated with initial and final states gives

$$T_1 + V(r_1) = T_2 + V(r_2)$$

This demonstrates that the total mechanical energy E, defined as the sum of kinetic and potential energy, remains constant throughout the motion:

$$E = T + V(r) = \text{constant}$$

2. Central Orbit: Equation of Motion in Plane Polar Coordinates

The path or trajectory traced out by a particle moving under the influence of a central force is called a central orbit. Since the motion is confined to a fixed plane, it is convenient to describe the trajectory using plane polar coordinates (r, θ) with the centre of force located at the origin O.

Kinematics in Plane Polar Coordinates

Let $\hat{r}$ be the unit vector along the radial direction and $\hat{\theta}$ be the unit vector perpendicular to $\hat{r}$ in the direction of increasing polar angle θ. These unit vectors can be expressed in terms of fixed Cartesian unit vectors $\hat{i}$ and $\hat{j}$ as

$$\hat{r} = \cos\theta \hat{i} + \sin\theta \hat{j}$$

$$\hat{\theta} = -\sin\theta \hat{i} + \cos\theta \hat{j}$$

Differentiating $\hat{r}$ with respect to time using the chain rule gives

$$\frac{d\hat{r}}{dt} = \frac{d\hat{r}}{d\theta}\frac{d\theta}{dt} = (-\sin\theta\hat{i} + \cos\theta\hat{j})\dot{\theta} = \dot{\theta}\hat{\theta}$$

Similarly, differentiating $\hat{\theta}$ with respect to time yields

$$\frac{d\hat{\theta}}{dt} = \frac{d\hat{\theta}}{d\theta}\frac{d\theta}{dt} = (-\cos\theta\hat{i} - \sin\theta\hat{j})\dot{\theta} = -\dot{\theta}\hat{r}$$

The position vector of the particle is written as

$$\vec{r} = r\hat{r}$$

Taking the time derivative of $\vec{r}$ to find the velocity vector $\vec{v}$ yields

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$$

where $v_r = \dot{r}$ is the radial component of velocity and $v_\theta = r\dot{\theta}$ is the transverse component of velocity.

Differentiating the velocity vector $\vec{v}$ with respect to time gives the acceleration vector $\vec{a}$:

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}\left(\dot{r}\hat{r} + r\dot{\theta}\hat{\theta}\right) = \ddot{r}\hat{r} + \dot{r}\frac{d\hat{r}}{dt} + (\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta} + r\dot{\theta}\frac{d\hat{\theta}}{dt}$$

Substituting the derivatives of the unit vectors $\frac{d\hat{r}}{dt} = \dot{\theta}\hat{\theta}$ and $\frac{d\hat{\theta}}{dt} = -\dot{\theta}\hat{r}$ into this expression gives

$$\vec{a} = \ddot{r}\hat{r} + \dot{r}(\dot{\theta}\hat{\theta}) + (\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta} + r\dot{\theta}(-\dot{\theta}\hat{r})$$

Grouping the components along $\hat{r}$ and $\hat{\theta}$ yields

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}$$

Here, the radial acceleration component is $a_r = \ddot{r} - r\dot{\theta}^2$ and the transverse acceleration component is $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$.

The transverse acceleration can be rewritten as a total time derivative:

$$a_\theta = \frac{1}{r}\frac{d}{dt}\left(r^2\dot{\theta}\right)$$

Expanding this derivative confirms the identity:

$$\frac{1}{r}\frac{d}{dt}\left(r^2\dot{\theta}\right) = \frac{1}{r}\left(2r\dot{r}\dot{\theta} + r^2\ddot{\theta}\right) = 2\dot{r}\dot{\theta} + r\ddot{\theta}$$

Equations of Motion

Applying Newton's second law $\vec{F} = m\vec{a}$ for a particle of mass m subject to a central force $\vec{F} = F(r)\hat{r} + 0\hat{\theta}$, we equate components in the radial and transverse directions:

1. Radial component:

$$m(\ddot{r} - r\dot{\theta}^2) = F(r)$$

1. Transverse component:

$$m\left(\frac{1}{r}\frac{d}{dt}\left(r^2\dot{\theta}\right)\right) = 0$$

Since $m \neq 0$ and $r \neq 0$, the transverse equation reduces to

$$\frac{1}{r}\frac{d}{dt}\left(r^2\dot{\theta}\right) = 0$$

Conservation of Angular Momentum

Multiplying the transverse equation of motion by r gives

$$\frac{d}{dt}\left(r^2\dot{\theta}\right) = 0$$

Integrating this equation with respect to time t yields

$$r^2\dot{\theta} = h$$

where h is a constant of integration representing the angular momentum per unit mass. Multiplying both sides by the mass m of the particle gives the magnitude of the orbital angular momentum l:

$$l = mr^2\dot{\theta} = \text{constant}$$

This confirms that the angular momentum l is an explicit constant of motion in a plane polar coordinate system.

3. Two-Body Central Force Problem: Reduction to One-Body Problem

Consider an isolated system composed of two point particles of masses m_1 and m_2 located at position vectors $\vec{r}_1$ and $\vec{r}_2$ respectively, measured from a fixed arbitrary origin O in an inertial frame of reference. Suppose these particles interact exclusively with each other through a central force acting along the line connecting their centers.

According to Newton's second law of motion, the equation of motion for the first particle of mass m_1 is written as:

$$m_1 \frac{d^2\vec{r}_1}{dt^2} = \vec{F}_{12}$$

where $\vec{F}_{12}$ denotes the force exerted on particle 1 by particle 2.

Similarly, the equation of motion for the second particle of mass m_2 is given by:

$$m_2 \frac{d^2\vec{r}_2}{dt^2} = \vec{F}_{21}$$

where $\vec{F}_{21}$ denotes the force exerted on particle 2 by particle 1.

By Newton's third law of motion, the mutual forces of interaction are equal in magnitude and opposite in direction:

$$\vec{F}_{21} = -\vec{F}_{12}$$

Since the interaction is a central force, the force vector acts along the direction of the relative position vector $\vec{r}$, defined by:

$$\vec{r} = \vec{r}_1 - \vec{r}_2$$

with magnitude $r = |\vec{r}| = |\vec{r}_1 - \vec{r}_2|$ and unit vector $\hat{r} = \vec{r}/r$. The force can thus be expressed as:

$$\vec{F}_{12} = f(r)\hat{r}$$

To decouple the complete motion of the two-particle system into centre of mass motion and relative motion, we define the position vector of the centre of mass $\vec{R}$ as:

$$\vec{R} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}$$

Adding the individual equations of motion for m_1 and m_2 yields:

$$m_1 \frac{d^2\vec{r}_1}{dt^2} + m_2 \frac{d^2\vec{r}_2}{dt^2} = \vec{F}_{12} + \vec{F}_{21}$$

Substituting the relation $\vec{F}_{21} = -\vec{F}_{12}$ into the right-hand side gives:

$$m_1 \frac{d^2\vec{r}_1}{dt^2} + m_2 \frac{d^2\vec{r}_2}{dt^2} = 0$$

Expressing the left-hand side in terms of the centre of mass coordinate $\vec{R}$ and total mass $M = m_1 + m_2$, we obtain:

$$M \frac{d^2\vec{R}}{dt^2} = 0$$

Integrating this equation once with respect to time shows that the velocity of the centre of mass is constant:

$$\frac{d\vec{R}}{dt} = \vec{V}_{\text{cm}} = \text{constant}$$

This demonstrates that the centre of mass of an isolated two-body system moves uniformly along a straight line, representing an unaccelerated inertial frame.

To determine the equation governing the relative motion, we divide the individual equations of motion by their respective masses. For particle 1, we write:

$$\frac{d^2\vec{r}_1}{dt^2} = \frac{\vec{F}_{12}}{m_1}$$

For particle 2, using $\vec{F}_{21} = -\vec{F}_{12}$, we write:

$$\frac{d^2\vec{r}_2}{dt^2} = -\frac{\vec{F}_{12}}{m_2}$$

Subtracting the acceleration equation of particle 2 from that of particle 1 yields:

$$\frac{d^2\vec{r}_1}{dt^2} - \frac{d^2\vec{r}_2}{dt^2} = \frac{\vec{F}_{12}}{m_1} - \left(-\frac{\vec{F}_{12}}{m_2}\right)$$

Factoring out the force vector $\vec{F}_{12}$ gives the second time derivative of the relative position vector $\vec{r}$:

$$\frac{d^2\vec{r}}{dt^2} = \left(\frac{1}{m_1} + \frac{1}{m_2}\right)\vec{F}_{12}$$

We define the reduced mass μ of the two-body system by the relation:

$$\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}$$

Taking the common denominator and inverting gives the reduced mass explicitly:

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

Substituting the definition of reduced mass μ into the relative equation of motion produces:

$$\frac{d^2\vec{r}}{dt^2} = \frac{1}{\mu}\vec{F}_{12}$$

Multiplying both sides by μ yields the final equation of motion for the relative coordinate:

$$\mu\frac{d^2\vec{r}}{dt^2} = f(r)\hat{r}$$

This result proves that the relative motion of two mutually interacting bodies of masses m_1 and m_2 is mathematically equivalent to the motion of a single fictitious particle of reduced mass μ moving under the influence of the same central force $f(r)\hat{r}$ directed towards a fixed center of force located at the origin of the relative coordinate system.

4. Deduction of Kepler's Laws of Planetary Motion

Kepler's Second Law (Law of Areas)

Kepler's second law states that the radius vector drawn from the Sun to a planet sweeps out equal areas in equal intervals of time; that is, the areal velocity of the planet is constant.

Consider a particle of reduced mass μ moving under a central force $\vec{F}(\vec{r}) = f(r)\hat{r}$. The torque $\vec{N}$ acting on the particle about the center of force is given by:

$$\vec{N} = \vec{r} \times \vec{F}(\vec{r})$$

Substituting the central force vector $\vec{F}(\vec{r}) = f(r)\hat{r}$ into the torque equation gives:

$$\vec{N} = \vec{r} \times [f(r)\hat{r}] = f(r)(\vec{r} \times \hat{r}) = 0$$

since the position vector $\vec{r}$ and the unit vector $\hat{r}$ are parallel.

Because torque is equal to the time rate of change of orbital angular momentum $\vec{L}$, we have:

$$\frac{d\vec{L}}{dt} = \vec{N} = 0$$

Integrating with respect to time shows that the angular momentum vector $\vec{L}$ is a constant of motion:

$$\vec{L} = \vec{r} \times \vec{p} = \mu\left(\vec{r} \times \frac{d\vec{r}}{dt}\right) = \text{constant vector}$$

Now consider an infinitesimal displacement $d\vec{r}$ of the planet along its trajectory in an infinitesimal time interval dt. The area dA swept out by the position vector $\vec{r}$ is given by half the area of the parallelogram formed by vectors $\vec{r}$ and $d\vec{r}$:

$$dA = \frac{1}{2}|\vec{r} \times d\vec{r}|$$

Dividing both sides by dt yields the rate at which area is swept out, known as the areal velocity:

$$\frac{dA}{dt} = \frac{1}{2}\left|\vec{r} \times \frac{d\vec{r}}{dt}\right|$$

Recalling the definition of angular momentum magnitude $L = \mu|\vec{r} \times \frac{d\vec{r}}{dt}|$, we express the vector product magnitude as:

$$\left|\vec{r} \times \frac{d\vec{r}}{dt}\right| = \frac{L}{\mu}$$

Substituting this relation into the expression for areal velocity gives:

$$\frac{dA}{dt} = \frac{L}{2\mu}$$

Since both the angular momentum magnitude L and the reduced mass μ are constants of motion, the areal velocity is strictly constant:

$$\frac{dA}{dt} = \text{constant}$$

This completes the deduction of Kepler's second law.

Kepler's First Law (Law of Ellipses)

Kepler's first law states that every planet moves in an elliptical orbit around the Sun, with the Sun situated at one of the two foci of the ellipse.

The gravitational attractive force between the Sun of mass M_s and a planet of mass m separated by distance r is an inverse-square central force:

$$f(r) = -\frac{GM_s m}{r^2} = -\frac{k}{r^2}$$

where $k = GM_s m$ is the gravitational coupling constant.

The differential equation for the orbit of a particle of reduced mass μ under a central force $f(r)$ in terms of the reciprocal variable $u(\theta) = 1/r(\theta)$ is given by:

$$\frac{d^2u}{d\theta^2} + u = -\frac{\mu}{L^2u^2}f(1/u)$$

Substituting the inverse-square force law $f(1/u) = -ku^2$ into the orbit equation yields:

$$\frac{d^2u}{d\theta^2} + u = -\frac{\mu}{L^2u^2}(-ku^2)$$

Simplifying the right-hand side gives:

$$\frac{d^2u}{d\theta^2} + u = \frac{\mu k}{L^2}$$

To solve this second-order differential equation, we introduce a substitute variable $w(\theta)$:

$$w(\theta) = u(\theta) - \frac{\mu k}{L^2}$$

Differentiating $w(\theta)$ twice with respect to θ gives:

$$\frac{d^2w}{d\theta^2} = \frac{d^2u}{d\theta^2}$$

Substituting w and its second derivative into the orbit differential equation transforms it into the simple harmonic oscillator equation:

$$\frac{d^2w}{d\theta^2} + w = 0$$

The general solution of this linear differential equation is:

$$w(\theta) = C\cos(\theta - \theta_0)$$

where C is a constant of integration related to orbital energy and θ_0 is an initial phase angle. Orienting the coordinate axes such that $\theta_0 = 0$, we have:

$$w(\theta) = C\cos\theta$$

Replacing $w(\theta)$ back in terms of $u(\theta)$ gives:

$$u(\theta) - \frac{\mu k}{L^2} = C\cos\theta$$

Solving for $u(\theta) = 1/r(\theta)$ yields:

$$u(\theta) = \frac{\mu k}{L^2} + C\cos\theta$$

Factoring out the term $\frac{\mu k}{L^2}$ on the right-hand side results in:

$$u(\theta) = \frac{\mu k}{L^2}\left(1 + \frac{CL^2}{\mu k}\cos\theta\right)$$

We define the semi-latus rectum p and eccentricity e of the orbit as:

$$p = \frac{L^2}{\mu k}$$

$$e = \frac{CL^2}{\mu k}$$

Substituting p and e into the equation for $u(\theta)$ gives:

$$\frac{1}{r(\theta)} = \frac{1}{p}(1 + e\cos\theta)$$

Taking the reciprocal yields the standard polar equation of a conic section with one focus at the origin:

$$r(\theta) = \frac{p}{1 + e\cos\theta}$$

For bound planetary motion, the orbit is closed and bounded, which corresponds to an eccentricity $e < 1$. In analytical geometry, a conic section with $e < 1$ represents an ellipse. The center of force (the Sun) resides at one focus of this ellipse, which proves Kepler's first law.

Kepler's Third Law (Law of Periods)

Kepler's third law states that the square of the orbital period T of any planet is directly proportional to the cube of the semi-major axis a of its elliptical orbit.

The total area A enclosed by an ellipse with semi-major axis a and semi-minor axis b is given by:

$$A = \pi ab$$

Since the areal velocity $\frac{dA}{dt} = \frac{L}{2\mu}$ is constant, integrating the area swept out over one full period of revolution T gives:

$$A = \int_0^T \frac{dA}{dt}dt = \frac{L}{2\mu}T$$

Equating the geometrical expression for the area of the ellipse to the integral result yields:

$$\pi ab = \frac{L}{2\mu}T$$

Solving this equation explicitly for the orbital period T gives:

$$T = \frac{2\mu\pi ab}{L}$$

Squaring both sides of the equation for T yields:

$$T^2 = \frac{4\pi^2\mu^2a^2b^2}{L^2}$$

From the geometry of ellipses, the semi-minor axis b, semi-major axis a, and semi-latus rectum p satisfy the identity:

$$b^2 = ap$$

Recalling the expression for the semi-latus rectum $p = \frac{L^2}{\mu k}$, we substitute p into the relation for b^2:

$$b^2 = a\left(\frac{L^2}{\mu k}\right)$$

Substituting this expression for b^2 back into the T^2 equation gives:

$$T^2 = \frac{4\pi^2\mu^2 a^2}{L^2}\left(\frac{aL^2}{\mu k}\right)$$

Canceling common factors L^2 and μ simplifies the expression to:

$$T^2 = \frac{4\pi^2\mu}{k}a^3$$

For a system consisting of a planet of mass m orbiting the Sun of mass M_s, the force constant is $k = GM_sm$ and the reduced mass is $\mu = \frac{mM_s}{m+M_s}$. The ratio $\frac{\mu}{k}$ is therefore:

$$\frac{\mu}{k} = \frac{\frac{mM_s}{m+M_s}}{GM_sm} = \frac{1}{G(M_s+m)}$$

Because the Sun's mass M_s is immensely larger than the planet's mass m ($M_s \gg m$), the sum $M_s + m \approx M_s$, so:

$$\frac{\mu}{k} \approx \frac{1}{GM_s}$$

Substituting this constant ratio into the formula for T^2 gives:

$$T^2 = \left(\frac{4\pi^2}{GM_s}\right)a^3$$

The term inside the parentheses is a constant depending only on the universal gravitational constant G and the mass of the Sun M_s. Consequently, we obtain:

$$T^2 \propto a^3$$

which completes the deduction of Kepler's third law.

5. Differential Equation of the Orbit

To determine the spatial trajectory or orbit of a particle of mass m moving under a central force $\mathbf{F}(\mathbf{r}) = F(r)\hat{\mathbf{r}}$, it is convenient to eliminate time t from the equations of motion and express the radial coordinate r directly as a function of the polar angle θ.

In a plane polar coordinate system (r, θ), the radial component of Newton's second law of motion is given by

$$m(\ddot{r} - r\dot{\theta}^2) = F(r)$$

The central force exerts no torque, leading to the conservation of angular momentum l. The magnitude of angular momentum is expressed as

$$l = mr^2\dot{\theta} = \text{constant}$$

We introduce the constant parameter h, defined as the angular momentum per unit mass

$$h = \frac{l}{m}$$

Substituting the definition of h into the angular momentum expression gives the angular velocity $\dot{\theta}$ in terms of r

$$\dot{\theta} = \frac{h}{r^2}$$

To simplify the differential equation of motion, we introduce a variable substitution by defining u as the reciprocal of the radial distance r

$$u = \frac{1}{r}$$

Expressing r in terms of u yields

$$r = \frac{1}{u}$$

We now transform the time derivatives $\frac{dr}{dt}$ and $\frac{d^2r}{dt^2}$ into derivatives with respect to θ. Using the chain rule of differentiation, the radial velocity $\dot{r}$ is written as

$$\frac{dr}{dt} = \frac{d}{dt}\left(\frac{1}{u}\right) = -\frac{1}{u^2}\frac{du}{dt} = -\frac{1}{u^2}\frac{du}{d\theta}\frac{d\theta}{dt}$$

Substituting $\frac{d\theta}{dt} = \dot{\theta} = hu^2$ into the expression for radial velocity gives

$$\frac{dr}{dt} = -\frac{1}{u^2}\frac{du}{d\theta}(hu^2) = -h\frac{du}{d\theta}$$

Differentiating $\frac{dr}{dt}$ once more with respect to time t to find the radial acceleration $\ddot{r}$, we apply the chain rule again

$$\frac{d^2r}{dt^2} = \frac{d}{dt}\left(-h\frac{du}{d\theta}\right) = -h\frac{d}{d\theta}\left(\frac{du}{d\theta}\right)\frac{d\theta}{dt}$$

Substituting $\frac{d\theta}{dt} = hu^2$ into the second derivative yields

$$\frac{d^2r}{dt^2} = -h\left(\frac{d^2u}{d\theta^2}\right)(hu^2) = -h^2u^2\frac{d^2u}{d\theta^2}$$

Next, we express the centrifugal acceleration term $r\dot{\theta}^2$ in terms of u and h

$$r\dot{\theta}^2 = \left(\frac{1}{u}\right)(hu^2)^2 = h^2u^3$$

We now substitute the expressions for $\ddot{r}$ and $r\dot{\theta}^2$ back into the radial equation of motion $m(\ddot{r} - r\dot{\theta}^2) = F(r)$

$$m\left(-h^2u^2\frac{d^2u}{d\theta^2} - h^2u^3\right) = F\left(\frac{1}{u}\right)$$

Factoring out the term $-mh^2u^2$ on the left-hand side gives

$$-mh^2u^2\left(\frac{d^2u}{d\theta^2} + u\right) = F\left(\frac{1}{u}\right)$$

Dividing both sides by $-mh^2u^2$ yields Binet's differential equation of the central orbit

$$\frac{d^2u}{d\theta^2} + u = -\frac{F\left(\frac{1}{u}\right)}{mh^2u^2}$$

We now apply Binet's formula to an attractive inverse-square force law, represented by

$$F(r) = -\frac{k}{r^2}$$

Expressed in terms of the variable u, this force law becomes

$$F\left(\frac{1}{u}\right) = -ku^2$$

Substituting $F(1/u) = -ku^2$ into Binet's differential equation gives

$$\frac{d^2u}{d\theta^2} + u = -\frac{-ku^2}{mh^2u^2}$$

Simplifying the right-hand side yields the linear inhomogeneous differential equation

$$\frac{d^2u}{d\theta^2} + u = \frac{k}{mh^2}$$

The general solution of this equation is the sum of the complementary function $u_c(\theta)$ and a particular integral u_p

$$u(\theta) = A\cos(\theta - \theta_0) + \frac{k}{mh^2}$$

Choosing the orientation of the polar axis such that $\theta_0 = 0$ aligns the reference line with the point of closest approach, simplifying the solution to

$$u(\theta) = \frac{k}{mh^2} + A\cos\theta$$

Factoring out the constant $\frac{k}{mh^2}$ gives

$$u(\theta) = \frac{k}{mh^2}\left(1 + \frac{Amh^2}{k}\cos\theta\right)$$

We define the dimensionless eccentricity e of the orbit as

$$e = \frac{Amh^2}{k}$$

Substituting e into the solution for $u(\theta)$ gives

$$u(\theta) = \frac{k}{mh^2}(1 + e\cos\theta)$$

Taking the reciprocal of $u(\theta)$ yields the orbital equation for $r(\theta)$

$$r(\theta) = \frac{\frac{mh^2}{k}}{1 + e\cos\theta}$$

Defining the semi-latus rectum parameter p as

$$p = \frac{mh^2}{k}$$

We arrive at the standard polar equation of a conic section with one focus at the origin

$$r(\theta) = \frac{p}{1 + e\cos\theta}$$

6. Effective Potential Energy and Radial Motion

The motion of a reduced mass μ under a conservative central force field $V(r)$ can be effectively analyzed using energy conservation principles, reducing the two-dimensional planar motion to an equivalent one-dimensional radial motion.

The total mechanical energy E of the system is the sum of its kinetic energy T and potential energy $V(r)$

$$E = T + V(r)$$

In plane polar coordinates, the kinetic energy T of the mass μ is written as

$$T = \frac{1}{2}\mu\left(\dot{r}^2 + r^2\dot{\theta}^2\right)$$

Thus, the total energy expression becomes

$$E = \frac{1}{2}\mu\dot{r}^2 + \frac{1}{2}\mu r^2\dot{\theta}^2 + V(r)$$

The orbital angular momentum l is a constant of motion given by

$$l = \mu r^2 \dot{\theta}$$

Solving for the angular velocity $\dot{\theta}$ in terms of l gives

$$\dot{\theta} = \frac{l}{\mu r^2}$$

Substituting this expression for $\dot{\theta}$ back into the total energy equation yields

$$E = \frac{1}{2}\mu \left(\frac{dr}{dt}\right)^2 + \frac{1}{2}\mu r^2 \left(\frac{l}{\mu r^2}\right)^2 + V(r)$$

Simplifying the second term gives

$$E = \frac{1}{2}\mu \left(\frac{dr}{dt}\right)^2 + \frac{l^2}{2\mu r^2} + V(r)$$

We group the terms dependent solely on position r by defining the effective potential energy $V_{\text{eff}}(r)$

$$V_{\text{eff}}(r) = V(r) + \frac{l^2}{2\mu r^2}$$

Here, the term $\frac{l^2}{2\mu r^2}$ is referred to as the centrifugal potential energy barrier. Using $V_{\text{eff}}(r)$, the total energy equation takes the form of a one-dimensional energy equation for the radial coordinate r

$$\frac{1}{2}\mu \left(\frac{dr}{dt}\right)^2 + V_{\text{eff}}(r) = E$$

For an attractive inverse-square force field, the actual potential energy is $V(r) = -\frac{k}{r}$ (where $k > 0$). Substituting $V(r)$ into $V_{\text{eff}}(r)$ gives

$$V_{\text{eff}}(r) = -\frac{k}{r} + \frac{l^2}{2\mu r^2}$$

Radial Motion Analysis from the Shape of $V_{\text{eff}}(r)$

The behavior of radial motion is governed by the relative magnitudes of the total energy E and the effective potential curve $V_{\text{eff}}(r)$.

1. **Behavior at the Boundaries:**

As $r \to 0$, the repulsive centrifugal barrier $+\frac{l^2}{2\mu r^2}$ dominates, causing $V_{\text{eff}}(r) \to +\infty$. This prevents any particle with non-zero angular momentum ($l \neq 0$) from reaching the center $r = 0$. As $r \to \infty$, both potential terms decay to zero, making $V_{\text{eff}}(r) \to 0$ from below.

1. **Minimum of the Effective Potential:**

The equilibrium position r_0 where the effective potential reaches its absolute minimum is found by setting the first derivative of $V_{\text{eff}}(r)$ with respect to r to zero

$$\frac{dV_{\text{eff}}}{dr} = \frac{k}{r^2} - \frac{l^2}{\mu r^3} = 0$$

Solving for r_0 gives the radius of a circular orbit

$$r_0 = \frac{l^2}{\mu k}$$

Evaluating $V_{\text{eff}}(r)$ at $r = r_0$ yields the absolute minimum effective potential energy E_{min}

$$V_{\text{eff}}(r_0) = -\frac{k}{\left(\frac{l^2}{\mu k}\right)} + \frac{l^2}{2\mu\left(\frac{l^2}{\mu k}\right)^2} = -\frac{\mu k^2}{l^2} + \frac{\mu k^2}{2l^2} = -\frac{\mu k^2}{2l^2} = E_{\text{min}}$$

1. **Turning Points, Allowed, and Forbidden Regions:**

Since kinetic energy $\frac{1}{2}\mu\dot{r}^2 = E - V_{\text{eff}}(r)$ must be non-negative (≥ 0), motion is physically possible only in regions where

$$E \geq V_{\text{eff}}(r)$$

These regions are the **allowed regions**. Regions where $E < V_{\text{eff}}(r)$ are **forbidden regions** because they would require imaginary radial velocities. Points where $E = V_{\text{eff}}(r)$ correspond to $\dot{r} = 0$; these are the radial **turning points** (r_{min} and r_{max}) where the radial velocity vanishes and reverses direction.

1. **Classification of Orbits based on Total Energy E:**

- $E < E_{\text{min}}$**:** Motion is physically impossible as $E < V_{\text{eff}}(r)$ everywhere. - $E = E_{\text{min}}$**:** The total energy equals the minimum of V_{eff}. There is only a single turning point at $r = r_0$. The radial velocity $\dot{r} = 0$ constantly, meaning $r(t) = r_0$ for all time. The orbit is a stable **bound circular orbit**. - $E_{\text{min}} < E < 0$**:** The energy line intersects $V_{\text{eff}}(r)$ at two distinct turning points, r_{min} (periapsis) and r_{max} (apoapsis). The radial distance is confined between $r_{\text{min}} \leq r \leq r_{\text{max}}$. The motion is **bound**, and the geometric shape of the orbit is an **ellipse**. - $E = 0$**:** There is a single inner turning point r_{min}, while $r_{\text{max}} \to \infty$. The particle can move to infinity, approaching it with zero residual radial kinetic energy. The motion is **unbound**, and the orbit is a **parabola**. - $E > 0$**:** There is one turning point of closest approach r_{min}. As $r \to \infty$, the particle retains a finite non-zero kinetic energy $\frac{1}{2}\mu\dot{r}_\infty^2 = E$. The motion is **unbound**, and the orbit is a **hyperbola**.

7. Stability of Circular Orbits

In a central force field, the radial motion of a particle of reduced mass μ is governed by an effective potential energy $V_{\text{eff}}(r)$ that incorporates both the actual potential energy $V(r)$ and a term arising from angular momentum conservation. The effective potential energy is defined as

$$V_{\text{eff}}(r) = V(r) + \frac{l^2}{2\mu r^2}$$

where l is the constant magnitude of the angular momentum of the system.

The equation of motion for the radial coordinate r is expressed in terms of this effective potential as

$$\mu \frac{d^2 r}{dt^2} = -\frac{dV_{\text{eff}}}{dr}$$

Differentiating the expression for the effective potential energy with respect to r gives

$$\frac{dV_{\text{eff}}}{dr} = \frac{dV}{dr} - \frac{l^2}{\mu r^3}$$

Using the relationship between central force $F(r)$ and potential energy $V(r)$, given by $F(r) = -\frac{dV}{dr}$, this radial derivative becomes

$$\frac{dV_{\text{eff}}}{dr} = -F(r) - \frac{l^2}{\mu r^3}$$

For a circular orbit, the radial distance remains constant at $r = r_0$, which means that the radial acceleration must be zero, $\frac{d^2 r}{dt^2} = 0$. Setting the equation of radial motion to zero gives the condition for a circular orbit as

$$\left.\frac{dV_{\text{eff}}}{dr}\right|_{r=r_0} = 0$$

Substituting the derivative of the effective potential into this equilibrium condition yields

$$-F(r_0) - \frac{l^2}{\mu r_0^3} = 0$$

Rearranging this equality shows that at $r = r_0$, the inward attractive central force balances the outward centrifugal force:

$$-F(r_0) = \frac{l^2}{\mu r_0^3}$$

To evaluate whether this circular orbit is stable against small radial perturbations, consider a small displacement $x(t)$ away from the equilibrium radius r_0, defined by

$$r(t) = r_0 + x(t)$$

Expanding the effective potential $V_{\text{eff}}(r)$ in a Taylor series about the equilibrium position $r = r_0$ gives

$$V_{\text{eff}}(r) = V_{\text{eff}}(r_0) + \left.\frac{dV_{\text{eff}}}{dr}\right|_{r=r_0} x + \frac{1}{2}\left.\frac{d^2 V_{\text{eff}}}{dr^2}\right|_{r=r_0} x^2 + \dots$$

Since the circular orbit condition requires $\left.\frac{dV_{\text{eff}}}{dr}\right|_{r=r_0} = 0$, the linear term vanishes, yielding the approximation

$$V_{\text{eff}}(r) \approx V_{\text{eff}}(r_0) + \frac{1}{2}\left.\frac{d^2V_{\text{eff}}}{dr^2}\right|_{r=r_0} x^2$$

Substituting this approximated potential into the radial equation of motion $\mu\frac{d^2x}{dt^2} = -\frac{dV_{\text{eff}}}{dx}$ gives

$$\mu\frac{d^2x}{dt^2} = -\frac{d}{dx}\left[V_{\text{eff}}(r_0) + \frac{1}{2}\left.\frac{d^2V_{\text{eff}}}{dr^2}\right|_{r=r_0} x^2\right]$$

Performing the differentiation with respect to x yields

$$\mu\frac{d^2x}{dt^2} = -\left.\frac{d^2V_{\text{eff}}}{dr^2}\right|_{r=r_0} x$$

For the particle to execute stable, bounded oscillations about r_0 rather than moving exponentially away, the restoring force must oppose the displacement x. This requires the second derivative of the effective potential at r_0 to be positive, giving the stability condition:

$$\left.\frac{d^2V_{\text{eff}}}{dr^2}\right|_{r=r_0} > 0$$

Now consider a central force obeying an attractive power-law of the form

$$F(r) = -kr^n$$

where $k > 0$ is a constant. The first derivative of the effective potential for this force law is

$$\frac{dV_{\text{eff}}}{dr} = kr^n - \frac{l^2}{\mu r^3}$$

Setting this derivative to zero at the equilibrium radius $r = r_0$ gives

$$kr_0^n - \frac{l^2}{\mu r_0^3} = 0$$

Solving this equation for the parameter $\frac{l^2}{\mu}$ yields

$$\frac{l^2}{\mu} = kr_0^{n+3}$$

Differentiating $\frac{dV_{\text{eff}}}{dr}$ a second time with respect to r gives

$$\frac{d^2V_{\text{eff}}}{dr^2} = \frac{d}{dr}\left(kr^n - \frac{l^2}{\mu r^3}\right)$$

Executing the derivative yields

$$\frac{d^2V_{\text{eff}}}{dr^2} = nkr^{n-1} + \frac{3l^2}{\mu r^4}$$

Evaluating this second derivative at $r = r_0$ gives

$$\left.\frac{d^2V_{\text{eff}}}{dr^2}\right|_{r=r_0} = nkr_0^{n-1} + \frac{3l^2}{\mu r_0^4}$$

Substituting the relation $\frac{l^2}{\mu} = kr_0^{n+3}$ into the second term of this derivative gives

$$\left.\frac{d^2V_{\text{eff}}}{dr^2}\right|_{r=r_0} = nkr_0^{n-1} + \frac{3(kr_0^{n+3})}{r_0^4}$$

Simplifying the exponent in the second term gives

$$\left.\frac{d^2V_{\text{eff}}}{dr^2}\right|_{r=r_0} = nkr_0^{n-1} + 3kr_0^{n-1}$$

Factoring out kr_0^{n-1} gives

$$\left.\frac{d^2V_{\text{eff}}}{dr^2}\right|_{r=r_0} = (n+3)kr_0^{n-1}$$

Applying the stability condition $\left.\frac{d^2V_{\text{eff}}}{dr^2}\right|_{r=r_0} > 0$ to this result leads to

$$(n+3)kr_0^{n-1} > 0$$

Since $k > 0$ and $r_0 > 0$, dividing both sides by kr_0^{n-1} yields

$$n + 3 > 0$$

Rearranging terms gives the condition for a stable circular orbit under a power-law force:

$$n > -3$$

For a stable circular orbit ($n > -3$), small radial perturbations result in simple harmonic motion about r_0. Dividing the linearized equation of motion $\mu\frac{d^2x}{dt^2} + \left.\frac{d^2V_{\text{eff}}}{dr^2}\right|_{r=r_0} x = 0$ by μ gives

$$\frac{d^2x}{dt^2} + \omega_r^2 x = 0$$

where the frequency of small radial oscillations ω_r is defined as

$$\omega_r = \sqrt{\frac{1}{\mu}\left.\frac{d^2V_{\text{eff}}}{dr^2}\right|_{r=r_0}}$$

Substituting $\left.\frac{d^2V_{\text{eff}}}{dr^2}\right|_{r=r_0} = (n+3)kr_0^{n-1}$ into the frequency expression gives

$$\omega_r = \sqrt{\frac{(n+3)kr_0^{n-1}}{\mu}}$$

Since the angular velocity of the circular orbit is $\omega_\theta = \frac{d\theta}{dt} = \frac{l}{\mu r_0^2}$, and using $\frac{l^2}{\mu^2 r_0^4} = \frac{kr_0^{n-1}}{\mu}$, the radial oscillation frequency can be written directly in terms of the orbital angular frequency:

$$\omega_r = \sqrt{n+3}\,\omega_\theta$$

8. Classification of Orbits

For an inverse-square central force field, such as gravitational attraction, the force magnitude is given by $F(r) = -\frac{k}{r^2}$ with $k > 0$. The corresponding central potential energy is

$$V(r) = -\frac{k}{r}$$

Solving the differential equation of the central orbit yields the standard polar equation of a conic section with one focus at the origin:

$$\frac{p}{r} = 1 + e\cos\theta$$

Here, p is the semilatus rectum defined by

$$p = \frac{l^2}{\mu k}$$

and e is the eccentricity of the orbit, which is determined by the total energy E and angular momentum l through the relation

$$e = \sqrt{1 + \frac{2El^2}{\mu k^2}}$$

The value of the total mechanical energy E determines the sign of the quantity $\frac{2El^2}{\mu k^2}$, which directly governs the magnitude of the eccentricity e and classifies the geometry of the orbit into distinct types:

For negative total energy ($E < 0$), the quantity under the square root satisfies $\frac{2El^2}{\mu k^2} < 0$, which implies

$$e < 1$$

This corresponds to a bound orbit where the particle remains confined within a finite radial region $r_{\min} \leq r \leq r_{\max}$.

When the energy attains its absolute minimum possible value for a given angular momentum l, defined by

$$E_{\min} = -\frac{\mu k^2}{2l^2}$$

substituting $E_{\min}$ into the eccentricity formula gives

$$e = \sqrt{1 + \frac{2\left(-\frac{\mu k^2}{2l^2}\right) l^2}{\mu k^2}}$$

Simplifying the term inside the square root gives

$$e = \sqrt{1 - 1} = 0$$

An eccentricity of $e = 0$ corresponds to a circular orbit of constant radius $r = p = \frac{l^2}{\mu k}$.

For energies in the range $E_{\min} < E < 0$, the eccentricity satisfies

$$0 < e < 1$$

This range of values represents an elliptical orbit. The point of closest approach (periapsis) and the point of farthest distance (apoapsis) are defined by $r_{\min} = \frac{p}{1+e}$ and $r_{\max} = \frac{p}{1-e}$, respectively.

For zero total energy ($E = 0$), substituting $E = 0$ into the expression for eccentricity yields

$$e = \sqrt{1 + 0} = 1$$

An eccentricity of $e = 1$ describes a parabolic orbit. This is an unbound trajectory in which the particle approaches from infinity with just enough energy to escape to infinity, coming to rest as $r \to \infty$.

For positive total energy ($E > 0$), the term $\frac{2El^2}{\mu k^2}$ is strictly positive, which yields

$$e > 1$$

An eccentricity of $e > 1$ corresponds to a hyperbolic orbit. This is an unbound trajectory in which the particle approaches from infinity, undergoes deflection by the central force, and escapes back to infinity with a non-zero residual velocity.

In summary, the total mechanical energy E and angular momentum l completely determine both the geometry and scale of a central orbit under an inverse-square force:

1. The total energy E specifies the geometric nature of the trajectory by fixing the eccentricity $e = \sqrt{1 + \frac{2El^2}{\mu k^2}}$, differentiating bound trajectories ($E < 0$, circle or ellipse) from unbound trajectories ($E \geq 0$, parabola or hyperbola).
2. The angular momentum l sets the spatial scale of the trajectory by determining the semilatus rectum $p = \frac{l^2}{\mu k}$. Together, E and l uniquely specify the orbital parameters and physical path of the particle.

Questions

Short form first (2–3 marks), then long form (5–10 marks). **No Assam University questions were found for this unit in the paper bank** — the questions below are from other universities' papers and standard syllabus-derived questions, marked as such.

Short-form questions

S1. [standard] Define a central force and state its properties.

Answer in brief: Section 1 — a force always directed towards or away from a fixed centre, with magnitude depending only on the distance from the centre; angular momentum about the centre is conserved and the motion is planar.

S2. [standard] Show that a central force is conservative.

Answer in brief: Section 1 — the work done between two points depends only on the radial distances, so a potential energy $V(r)$ exists with $F(r) = -dV/dr$ and total mechanical energy is conserved.

S3. [standard] State Kepler's laws of planetary motion.

Answer in brief: Section 4 — (1) planets move in ellipses with the Sun at one focus; (2) the radius vector sweeps out equal areas in equal times; (3) T^2 is proportional to a^3.

S4. [standard] What is the effective potential? Write it for the inverse-square force.

Answer in brief: Section 6 — $V_{eff}(r) = V(r) + l^2/(2\mu r^2)$; for the inverse-square force $V_{eff}(r) = -k/r + l^2/(2\mu r^2)$.

S5. [standard] State the condition for stability of a circular orbit.

Answer in brief: Section 7 — the circular orbit sits at a minimum of the effective potential, $dV_{eff}/dr = 0$ and $d^2V_{eff}/dr^2 > 0$ at r_0; for a power-law force $F = -kr^n$ this needs $n > -3$.

S6. [standard] How does the total energy classify the orbit of a particle under an inverse-square force?

Answer in brief: Section 8 — $E < 0$ ellipse (bound), $E = 0$ parabola, $E > 0$ hyperbola (unbound); $e = 0$ is a circle.

Long-form questions

L1. [non-AU] State Kepler's laws of planetary motion and derive the first law of planetary motion. — CBCS-pattern UG Physics-I, Mechanics, Waves and Oscillations, Section B.

What to write: Section 4 in full — state all three laws, then derive the first law from the orbit equation for the inverse-square force: $u = (1/l)(1 + e\cos\theta)$ is a conic, and for $E < 0$ the eccentricity $e < 1$, so the orbit is an ellipse with the Sun at one focus.

L2. [non-AU] Obtain Kepler's second law of planetary motion. — St. Aloysius, M.Sc Sem-I, CORE PH010102 Classical Mechanics, Nov 2021, Part B Q15.

What to write: Sections 1–2 and 4 — torque about the centre is zero, so angular momentum $l = mr^2\dot\theta$ is constant; the areal velocity $dA/dt = l/(2m)$ is therefore constant, which is the law of areas.

L3. [non-AU] How will we reduce a two-body problem to a one body problem? Derive the equation of motion of a particle moving under a central force. — St. Aloysius Nov 2021 Part A Q7; CBCS-pattern UG Physics-I Section B.

What to write: Sections 2–3 in full — separate the centre-of-mass motion, introduce the reduced mass $\mu = m_1 m_2/(m_1 + m_2)$, and derive the plane polar equations $m(\ddot{r} - r\dot{\theta}^2) = F(r)$ and $(1/r)d/dt(r^2\dot{\theta}) = 0$.

L4. [standard] Derive the differential equation of the orbit in plane polar coordinates for a particle under a central force, obtain the effective potential, and discuss radial motion, the condition for stability of circular orbits, and classification of orbits.

What to write: Sections 5–8 in full — the Binet formula and its inverse-square solution; $V_{eff}(r)$, turning points and allowed regions; the stability condition $d^2V_{eff}/dr^2 > 0$; orbit classification by E and e.

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.